Quantum Computing

Composite Systems and Entanglement

Lecture 2 Notes

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About these notes

These notes accompany a nine-lecture course on quantum computing designed and produced by the Lebanese Science Organization. Each chapter corresponds to one lecture. The lectures are concept-focused: they explain what a quantum state is, how entanglement, interference, and measurement combine to produce computation, and what is required to make that computation reliable. Linear algebra, complex numbers, and elementary probability are assumed.

Each chapter names its lecturer and date, opens with a short preface, develops its material with the derivations in place, and closes with a summary. Worked examples appear in numbered boxes, as do definitions, theorems, remarks, and common misconceptions. Works cited in the text are collected in a single reference list at the end of the document.

Conventions used throughout: qubits in a register are labelled from left to right, so that in $|ab\rangle$ the first qubit is a; the first qubit is the top wire in every circuit diagram; the computational basis of two qubits is ordered $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; circuits are read from left to right, so a gate U_1 followed by a gate U_2 acts as the matrix product U_2U_1.

This chapter is exported individually from the full lecture notes (2 of 9). These notes are made available for open, public study. Reuse conditions will be stated separately by the Lebanese Science Organization.

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Chapter 1

Composite Systems and Entanglement

Lecturer: Marc Fares

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The state of two qubits is a single vector in a four-dimensional space, not a pair of single-qubit vectors. This lecture builds that space with the tensor product, asks when a two-qubit state can be read as one state for each qubit, and studies what follows, physically and operationally, when it cannot. The Bell states are the running example: their joint outcomes are perfectly correlated while each qubit alone is completely random, no message can be sent through them, and the CHSH inequality shows that no local classical model reproduces their statistics.

1.1 The tensor product

In **??** a qubit was a unit vector in $\mathbb{C}^2$. A register of two qubits is not described by two such vectors side by side. It is described by one unit vector in a four-dimensional space whose basis vectors are labelled by the four classical two-bit strings. The construction that produces this space from the two single-qubit spaces is the tensor product.

Definition 1.1: Tensor product of qubit states

The state space of two qubits is $\mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$, with computational basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, in this order, where $|jk\rangle$ abbreviates $|j\rangle \otimes |k\rangle$. The first label refers to the first qubit, which is the top wire in a circuit. For single-qubit states $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and $|\phi\rangle = \gamma|0\rangle + \delta|1\rangle$ the product state is

$$|\psi\rangle \otimes |\phi\rangle = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle. \tag{1.1}$$

The tensor product is bilinear, and inner products factor: $(\langle\psi'|\otimes\langle\phi'|)(|\psi\rangle\otimes|\phi\rangle) = \langle\psi'|\psi\rangle\langle\phi'|\phi\rangle$. Operators compose in the same way: $(A \otimes B)(|\psi\rangle \otimes |\phi\rangle) = A|\psi\rangle \otimes B|\phi\rangle$. The coordinate form of these rules, the Kronecker product, is recorded below.

Coordinates. In the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ of Definition 1.1, the product state has the column vector

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix} \otimes \begin{pmatrix}\gamma\\ \delta\end{pmatrix} = \begin{pmatrix}\alpha\gamma\\ \alpha\delta\\ \beta\gamma\\ \beta\delta\end{pmatrix}, \tag{1.2}$$

that is, each entry of the first vector is replaced by that entry times the whole second vector. The same rule, the Kronecker product, gives the matrix of $A \otimes B$: each entry A_{jk} of the 2×2 matrix A is replaced by the 2×2 block $A_{jk}B$, producing a 4×4 matrix whose rows and columns are indexed by the two-bit strings in the same order. With this convention $(A \otimes B)(|\psi\rangle \otimes |\phi\rangle) = A|\psi\rangle \otimes B|\phi\rangle$ holds entrywise, and the matrix of $H \otimes \mathbb{I}$ displayed in section 1.1.2 is an instance.

We write $|\psi\rangle|\phi\rangle$ or $|\psi\phi\rangle$ for $|\psi\rangle \otimes |\phi\rangle$ when no confusion is possible. A gate acting on the first qubit alone is $U \otimes \mathbb{I}$, and a gate acting on the second alone is $\mathbb{I} \otimes U$. The general two-qubit state is any unit vector

$$|\Psi\rangle = \sum_{j,k\in\{0,1\}} c_{jk}\,|jk\rangle\,, \qquad \sum_{j,k} |c_{jk}|^2 = 1, \tag{1.3}$$

and measuring both qubits in the computational basis gives the string jk with probability $|c_{jk}|^2$, exactly as the Born rule prescribes in four dimensions.

1.1.1 Product states and entangled states

The vectors of the form $|\psi\rangle \otimes |\phi\rangle$ do not exhaust $\mathbb{C}^4$. A product state has four amplitudes that are all determined by the two pairs (α, β) and (γ, δ), while a general vector (1.3) has four independent amplitudes. Most two-qubit states are therefore not product states.

Definition 1.2: Product and entangled states

A two-qubit pure state $|\Psi\rangle$ is a product state if $|\Psi\rangle = |\psi\rangle \otimes |\phi\rangle$ for some single-qubit states $|\psi\rangle$ and $|\phi\rangle$. Otherwise it is entangled. In a product state, each qubit has a state of its own; in an entangled state, neither qubit has a pure state of its own, and the vector $|\Psi\rangle$ describes only the pair.

The canonical entangled state of this course is

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right). \tag{1.4}$$

It is not a product state: a product state with nonzero amplitudes on $|00\rangle$ and $|11\rangle$ must have $\alpha, \beta, \gamma, \delta$ all nonzero, and then its amplitude $\alpha\delta$ on $|01\rangle$ cannot vanish. In detail, suppose $|\Phi^+\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle) = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$. Comparing amplitudes, $\alpha\gamma = \beta\delta = 1/\sqrt{2}$ and $\alpha\delta = \beta\gamma = 0$. From $\alpha\gamma \neq 0$ both α and γ are nonzero; then $\alpha\delta = 0$ forces $\delta = 0$, which contradicts $\beta\delta \neq 0$. No such factorization exists. The same argument applies to all four Bell states, and in general the state (1.3) is a product state if and only if $c_{00}c_{11} = c_{01}c_{10}$, since a rank-one 2×2 matrix (c_{jk}) is exactly one with vanishing determinant.

The word "entangled" is thus a statement about the algebraic form of a vector. Its physical content is the subject of the next two sections.

1.1.2 The CNOT gate and Bell states

Single-qubit gates applied to the two qubits separately produce $U \otimes V$, which maps product states to product states. To create entanglement from a product input, a gate must act jointly on both qubits. The canonical example is the controlled-NOT gate, which flips the second qubit (the target) exactly when the first qubit (the control) is $|1\rangle$:

$$\mathrm{CNOT}|00\rangle = |00\rangle\,, \quad \mathrm{CNOT}|01\rangle = |01\rangle\,, \quad \mathrm{CNOT}|10\rangle = |11\rangle\,, \quad \mathrm{CNOT}|11\rangle = |10\rangle\,. \tag{1.5}$$

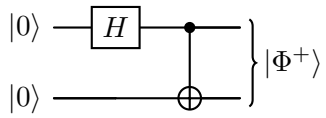
This defines a permutation of the four basis vectors, so CNOT is unitary and is its own inverse. On a basis state it computes the classical reversible operation $(j, k) \mapsto (j, j \oplus k)$. Its power comes from

linearity: applied to a superposition of control values, it produces a superposition of "flipped" and "not flipped" branches, which is precisely what a product state cannot express.

Start from $|00\rangle$, apply H to the first qubit to obtain $|+\rangle|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$, then apply CNOT. The branch $|00\rangle$ is unchanged and the branch $|10\rangle$ becomes $|11\rangle$, so

$$\mathrm{CNOT}\,(H \otimes \mathbb{I})\,|00\rangle = \frac{1}{\sqrt{2}}\,(|00\rangle + |11\rangle) = \left|\Phi^+\right\rangle. \tag{1.6}$$

Two gates suffice to turn a product state into an entangled one. Since circuits are read left to right and matrices act on the right, the circuit "H on the top wire, then CNOT" is the operator product $\mathrm{CNOT}\,(H \otimes \mathbb{I})$; **??** develops the circuit model systematically. The circuit is drawn as follows, with the control on the top wire and the target on the bottom wire.



In the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, the action (1.5) gives the matrices

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad H \otimes \mathbb{I} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}, \tag{1.7}$$

the second being the Kronecker product of H with the 2×2 identity: each entry H_{jk} is replaced by the block $H_{jk}\mathbb{I}$. Then

$$(H \otimes \mathbb{I})\,|00\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathrm{CNOT}\,(H \otimes \mathbb{I})\,|00\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \left|\Phi^+\right\rangle. \tag{1.8}$$

1.1.3 Exponential dimension

For n qubits the state space is $(\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}$, with one basis vector per n-bit string. A state of n qubits is specified by 2^n complex amplitudes; for $n = 300$ this exceeds the number of atoms in the observable universe. It is tempting to conclude that a quantum register "stores" or "processes" exponentially much information. That conclusion is false in the sense that matters.

> **Remark 1.1** Amplitudes are not readable
>
> Measuring an n-qubit register in the computational basis returns one n-bit string, chosen at random according to the 2^n probabilities $|c_x|^2$, and destroys the superposition. A single run reveals n bits, not 2^n amplitudes. Even a single probability $|c_x|^2$ can only be estimated from many identically prepared copies, at a cost quantified below, and the phases of the amplitudes are invisible to this measurement altogether. The exponential dimension is real, and it is what a classical simulation must pay for. It is not a storage capacity available to the user. Every algorithm in this course will have to arrange, through interference, that the one string we do read out carries the answer.

The cost of estimating one probability, mentioned in Remark 1.1, is the usual statistical one. Measuring N identically prepared copies of an n-qubit register in the computational basis and recording how often the string x appears gives an estimate of $p_x = |c_x|^2$ whose standard deviation is

$\sqrt{p_x(1-p_x)/N} \leqslant 1/(2\sqrt{N})$. To pin p_x down to precision ϵ therefore requires N of the order of $1/\epsilon^2$ copies, and this determines only the modulus of c_x, never its phase. Reading out all 2^n probabilities to fixed precision would require exponentially many copies, which is the quantitative content of the statement that the amplitudes are not a storage capacity available to the user.

1.2 Bell states and conditional states

The state $|\Phi^+\rangle$ is one of four mutually orthogonal two-qubit states, which together form a basis of $\mathbb{C}^4$. Each of them is maximally entangled, meaning that each qubit on its own is completely random in every basis, as Proposition 1.1 below makes precise for $|\Phi^+\rangle$.

Definition 1.3: Bell states

The Bell states are

$$\begin{aligned} |\Phi^+\rangle &= \frac{|00\rangle + |11\rangle}{\sqrt{2}}, & |\Phi^-\rangle &= \frac{|00\rangle - |11\rangle}{\sqrt{2}}, \\ |\Psi^+\rangle &= \frac{|01\rangle + |10\rangle}{\sqrt{2}}, & |\Psi^-\rangle &= \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \end{aligned} \tag{1.9}$$

They are orthonormal, so they form a basis, the Bell basis. The circuit of (1.6) maps the four computational basis states to the four Bell states, in the order $|00\rangle \mapsto |\Phi^+\rangle$, $|01\rangle \mapsto |\Psi^+\rangle$, $|10\rangle \mapsto |\Phi^-\rangle$, $|11\rangle \mapsto |\Psi^-\rangle$. The state $|\Psi^-\rangle$ is also called the singlet.

The circuit of (1.6) on the other three inputs gives $|01\rangle \mapsto |+\rangle\,|1\rangle \mapsto \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = |\Psi^+\rangle$, then $|10\rangle \mapsto |-\rangle\,|0\rangle \mapsto \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |\Phi^-\rangle$, and $|11\rangle \mapsto |-\rangle\,|1\rangle \mapsto \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) = |\Psi^-\rangle$. Since a unitary maps an orthonormal basis to an orthonormal basis, this also proves that the four Bell states are orthonormal. Running the circuit backwards, $(H \otimes \mathbb{I})\,\mathrm{CNOT}$, maps the Bell basis to the computational basis, so a measurement in the Bell basis can be performed by applying these two gates and measuring both qubits in the computational basis. This is used in **??**.

1.2.1 Joint probabilities and conditional states

Suppose two parties, conventionally Alice and Bob, each hold one qubit of a pair in the state (1.3). If both measure in the computational basis, the pair of outcomes jk occurs with probability $|c_{jk}|^2$. For $|\Phi^+\rangle$ this gives 00 and 11 with probability $\frac{1}{2}$ each and never 01 or 10: the outcomes are random but perfectly correlated.

Suppose instead that only Alice measures. Her outcome j occurs with probability $p_j = \sum_k |c_{jk}|^2$, and after she observes j the state of the pair is $|j\rangle \otimes |\phi_j\rangle$ with

$$|\phi_j\rangle = \frac{1}{\sqrt{p_j}} \sum_k c_{jk}\,|k\rangle\,. \tag{1.10}$$

This is the projection rule of **??** applied to the first factor only. The vector $|\phi_j\rangle$ is Bob's conditional state given Alice's outcome. For $|\Phi^+\rangle$ and a computational-basis measurement by Alice, $|\phi_0\rangle = |0\rangle$ and $|\phi_1\rangle = |1\rangle$: Bob's qubit is now in a definite basis state, the same as Alice's outcome. Before Alice measured, Bob's qubit had no pure state of its own; after she measures, it has one, though Bob does not know which until he is told.

The same holds in other bases, and this is what distinguishes $|\Phi^+\rangle$ from a pair of matching

classical bits. Using $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$,

$$|00\rangle + |11\rangle = \frac{1}{2}\left[(|+\rangle + |-\rangle) \otimes (|+\rangle + |-\rangle) + (|+\rangle - |-\rangle) \otimes (|+\rangle - |-\rangle)\right] = |++\rangle + |--\rangle\,, \quad (1.11)$$

because the cross terms $|+-\rangle$ and $|-+\rangle$ appear with opposite signs in the two products and cancel. Hence $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$ as well. If Alice measures in the $\{|+\rangle, |-\rangle\}$ basis, she obtains $+$ or $-$ with probability $\frac{1}{2}$ each, and Bob's conditional state is $|+\rangle$ or $|-\rangle$ accordingly. The same substitution with a minus sign gives $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle)$.

More generally, for any basis $\{|u\rangle, |v\rangle\}$ with real amplitudes, $|u\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle$ and $|v\rangle = -\sin\frac{\theta}{2}|0\rangle + \cos\frac{\theta}{2}|1\rangle$, a direct expansion gives $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|uu\rangle + |vv\rangle)$. This is why Bob's conditional state is exactly the basis vector Alice found, for every real basis. For a basis with complex amplitudes the conditional state is the complex conjugate of Alice's outcome, a distinction that plays no role in this course.

1.2.2 Entanglement versus classical correlation

Consider a classically correlated preparation: a referee flips a fair coin and prepares the pair in $|00\rangle$ or $|11\rangle$ according to the result, without telling Alice or Bob which. Measured in the computational basis, this preparation gives 00 and 11 with probability $\frac{1}{2}$ each, exactly like $|\Phi^+\rangle$. The two situations are distinguished only by measuring in another basis.

Example 1.1 (Bell state versus classical correlation)

Both qubits are measured in the $\{|+\rangle, |-\rangle\}$ basis. The first symbol of each outcome refers to Alice's qubit.

Outcome	$\|\Phi^+\rangle$	Coin-flip mixture of $\|00\rangle$ and $\|11\rangle$
$++$	$\frac{1}{2}$	$\frac{1}{4}$
$+-$	0	$\frac{1}{4}$
$-+$	0	$\frac{1}{4}$
$--$	$\frac{1}{2}$	$\frac{1}{4}$

The Bell state remains perfectly correlated in the new basis, because $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$. The classical mixture loses all correlation, because each of $|00\rangle$ and $|11\rangle$ gives the four outcomes with probability $\frac{1}{4}$ when both qubits are measured in the $\{|+\rangle, |-\rangle\}$ basis.

The first column of the table is the expansion $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$ derived above. For the second column, $|00\rangle$ measured in the $\{|+\rangle, |-\rangle\}$ basis on both qubits gives each outcome with probability $|\langle\pm|0\rangle|^2|\langle\pm|0\rangle|^2 = \frac{1}{4}$, and likewise $|11\rangle$; averaging over the coin leaves $\frac{1}{4}$ for each outcome.

The table is the two-qubit version of the argument in **??** that a superposition is not classical ignorance. A pair of matching classical bits is correlated in one basis. The state $|\Phi^+\rangle$ is correlated in every basis with real amplitudes, and the correlation is carried by the relative sign between $|00\rangle$ and $|11\rangle$, which the coin-flip mixture does not possess. Replacing that sign by a minus gives $|\Phi^-\rangle$, which is anticorrelated in the $\{|+\rangle, |-\rangle\}$ basis while remaining correlated in the computational basis.

Wrong Concept 1.1: Entangled qubits are just correlated coins

Two coins glued so that they always land the same way reproduce the computational-basis statistics of $|\Phi^+\rangle$. They do not reproduce Example 1.1: no way of preparing one classical bit per qubit in advance gives perfect correlation both in the computational basis and in

the $\{|+\rangle, |-\rangle\}$ basis, because a classical bit has one value, not one value per basis. A more elaborate classical record, with one prearranged answer for each basis, does reproduce that particular table; ruling out every such record is exactly what section 1.3 does, quantitatively, with measurement settings that are not restricted to two bases.

1.2.3 No-signalling

Alice's measurement changes Bob's conditional state, whatever the distance between them. It is natural to ask whether Alice can use this to send Bob a message. She cannot, and the reason is not a practical limitation but a theorem.

Proposition 1.1 No signalling

Let Alice and Bob share any two-qubit state. The probability distribution of Bob's outcome, in any basis he chooses, does not depend on which basis Alice measures in, nor on whether she measures at all. For $|\Phi^+\rangle$ in particular, Bob obtains each of his two outcomes with probability $\frac{1}{2}$ in every basis, regardless of Alice's actions.

The mechanism is simple. Bob's conditional state does depend on Alice's outcome, but Alice does not control her outcome; it is random, with probabilities fixed by the state. Averaged over her outcomes, which is all Bob has access to without a classical message from her, Bob's statistics are the same as if Alice had done nothing. The calculation rests on nothing more than the completeness relation $|u\rangle\langle u| + |v\rangle\langle v| = \mathbb{I}$ for Alice's basis.

Proof. Let the shared state be $|\Psi\rangle$, let Alice measure in the orthonormal basis $\{|u\rangle, |v\rangle\}$, and let Bob measure in the basis $\{|w\rangle, |w^\perp\rangle\}$. The probability that Bob obtains w is the sum over Alice's two possible outcomes of the joint probabilities,

$$\begin{aligned} P(w) &= |(\langle u| \otimes \langle w|)\,|\Psi\rangle|^2 + |(\langle v| \otimes \langle w|)\,|\Psi\rangle|^2 \\ &= \langle\Psi|\,[(|u\rangle\langle u| + |v\rangle\langle v|) \otimes |w\rangle\langle w|]\,|\Psi\rangle = \langle\Psi|\,(\mathbb{I} \otimes |w\rangle\langle w|)\,|\Psi\rangle\,. \end{aligned} \tag{1.12}$$

The last expression does not involve $|u\rangle$ or $|v\rangle$, so Bob's distribution is independent of Alice's choice of basis, and it coincides with what Bob would obtain if Alice did not measure. This proves Proposition 1.1. For $|\Phi^+\rangle$ specifically, the cross terms between $|00\rangle$ and $|11\rangle$ vanish because $\langle 0|1\rangle = 0$ on Alice's side, and

$$P(w) = \frac{1}{2}\left(|\langle w|0\rangle|^2 + |\langle w|1\rangle|^2\right) = \frac{1}{2}\,\langle w|w\rangle = \frac{1}{2} \tag{1.13}$$

for every unit vector $|w\rangle$. Bob's qubit behaves as a fair coin in every basis. ∎

Wrong Concept 1.2: Entanglement lets you send messages faster than light

The intuition is that Alice, by choosing a basis, "collapses" Bob's qubit into a state of her choosing. But she chooses only the basis, not the outcome, and the two conditional states occur with probabilities that leave Bob's marginal distribution unchanged. Bob sees a fair coin whatever Alice does. Only when Alice later tells Bob her outcome, over an ordinary channel, can he see the correlation, and by then the message has travelled no faster than that channel. Entanglement produces correlation, not communication; **??** will show that even teleportation needs two classical bits per qubit for exactly this reason.

1.3 Bell's theorem

Example 1.1 shows that one particular classical preparation fails to reproduce the statistics of $|\Phi^+\rangle$. A sceptic may reply that a cleverer classical model, in which each qubit carries a list of predetermined answers for every possible basis, would succeed. Bell's theorem [1], in the form due to Clauser, Horne, Shimony, and Holt (CHSH) [2], shows that no such model can reproduce the predictions of quantum mechanics, provided the model satisfies assumptions that we now make explicit.

1.3.1 The CHSH inequality

Definition 1.4: Local hidden-variable model

Alice chooses one of two measurement settings, a or a', and Bob one of two settings, b or b'; each measurement has outcomes ± 1. A local hidden-variable model for the experiment consists of a variable λ with a probability distribution $p(\lambda)$, fixed when the pair is prepared, and response functions $A(a,\lambda), A(a',\lambda), B(b,\lambda), B(b',\lambda) \in \{+1,-1\}$ such that the expectation of the product of outcomes is

$$E(a,b) = \sum_{\lambda} p(\lambda)\, A(a,\lambda)\, B(b,\lambda), \tag{1.14}$$

and similarly for the other three setting pairs. The model assumes locality (Alice's outcome depends on her setting and λ, not on Bob's setting, and conversely), determinism given λ (each outcome is a function of the local setting and λ), and measurement independence (the distribution $p(\lambda)$ does not depend on the settings, which are chosen freely).

The variable λ is unrestricted: it may be a pair of bits, a list of answers to every question, or any object whatever. Determinism given λ costs nothing, since any local randomness in the responses can be absorbed into λ. What the model forbids is any dependence of one party's outcome on the other party's choice of setting. The glued coins of the box above are a special case with λ equal to the shared coin value.

Consider the combination

$$S = E(a,b) + E(a,b') + E(a',b) - E(a',b'). \tag{1.15}$$

Theorem 1.1 CHSH inequality

In every local hidden-variable model, $|S| \leqslant 2$.

The proof is short enough to state in one line. For each fixed λ the four responses are definite numbers ± 1, and $S_\lambda = A(B+B') + A'(B-B')$. One of $B+B'$ and $B-B'$ is zero and the other is ± 2, so $S_\lambda = \pm 2$; averaging over λ gives $|S| \leqslant 2$. The bound is a statement about correlations between four quantities that possess values simultaneously, even though only two are measured on any given pair. The details follow.

Proof. Fix λ and write $A = A(a,\lambda)$, $A' = A(a',\lambda)$, $B = B(b,\lambda)$, $B' = B(b',\lambda)$, each equal to ± 1. The four possible value pairs (B, B') are the four deterministic strategies available to Bob for a

given λ, and for each of them

$$S_\lambda = AB + AB' + A'B - A'B' = A(B+B') + A'(B-B') = \begin{cases} 2A & (B,B') = (+1,+1), \\ 2A' & (B,B') = (+1,-1), \\ -2A' & (B,B') = (-1,+1), \\ -2A & (B,B') = (-1,-1). \end{cases} \tag{1.16}$$

In every case $S_\lambda = \pm 2$. Averaging with the weights $p(\lambda)$,

$$|S| = \Big| \sum_\lambda p(\lambda)\, S_\lambda \Big| \leqslant \sum_\lambda p(\lambda)\, |S_\lambda| = 2, \tag{1.17}$$

which is Theorem 1.1. The step that uses the assumptions is the first line: writing S_λ as a combination of four numbers A, A', B, B' that exist simultaneously and do not depend on the other party's setting. If A were allowed to depend on whether Bob chose b or b', the factorization $A(B+B')$ would not be available and the bound would fail. ■

1.3.2 The quantum correlator

Now let Alice and Bob share $|\Phi^+\rangle$ and measure along directions in the x-z plane of the Bloch sphere: the measurement at angle θ is the real basis $\{|u_\theta\rangle, |v_\theta\rangle\}$ with $|u_\theta\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle$, outcome $+1$ for $|u_\theta\rangle$ and -1 for $|v_\theta\rangle$, so that $\theta = 0$ is the computational basis and $\theta = \pi/2$ is the $\{|+\rangle, |-\rangle\}$ basis. A direct calculation, carried out below, gives

$$E(\theta_a, \theta_b) = \cos(\theta_a - \theta_b) \qquad \text{for } \big|\Phi^+\big\rangle, \tag{1.18}$$

so the correlation depends only on the angle between the two measurement directions, and equals 1 when they coincide, which recovers the perfect correlations of Example 1.1. With the settings $\theta_a = 0$, $\theta_{a'} = \pi/2$, $\theta_b = \pi/4$, $\theta_{b'} = -\pi/4$, each of the four correlators in (1.15) has modulus $1/\sqrt{2}$ and the signs conspire so that $S = 2\sqrt{2} > 2$. Quantum mechanics predicts a value that no local hidden-variable model can produce. The value $2\sqrt{2}$ is in fact the largest that any quantum state and any measurements can give; this is Tsirelson's bound, which we state without proof.

We now derive (1.18). Measuring at angle θ with outcomes ± 1 assigned to $|u_\theta\rangle$ and $|v_\theta\rangle$ is the same as measuring the observable $M(\theta) = |u_\theta\rangle\langle u_\theta| - |v_\theta\rangle\langle v_\theta| = \cos\theta\, Z + \sin\theta\, X$, whose eigenvectors are $|u_\theta\rangle$ and $|v_\theta\rangle$ with eigenvalues $+1$ and -1; this is checked by direct multiplication using the half-angle formulas. The expectation of the product of outcomes is

$$E(\theta_a, \theta_b) = \big\langle \Phi^+ \big|\, M(\theta_a) \otimes M(\theta_b)\, \big|\Phi^+\big\rangle, \tag{1.19}$$

and by bilinearity it suffices to know four expectations. From $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$, $X|0\rangle = |1\rangle$, $X|1\rangle = |0\rangle$,

$$\begin{aligned} (Z\otimes Z)\big|\Phi^+\big\rangle &= \big|\Phi^+\big\rangle, & (X\otimes X)\big|\Phi^+\big\rangle &= \big|\Phi^+\big\rangle, \\ (Z\otimes X)\big|\Phi^+\big\rangle &= \big|\Psi^-\big\rangle, & (X\otimes Z)\big|\Phi^+\big\rangle &= -\big|\Psi^-\big\rangle, \end{aligned} \tag{1.20}$$

so $\langle\Phi^+|\, Z\otimes Z\,|\Phi^+\rangle = \langle\Phi^+|\, X\otimes X\,|\Phi^+\rangle = 1$ and the two mixed expectations vanish by orthogonality of $|\Phi^+\rangle$ and $|\Psi^-\rangle$. Therefore

$$E(\theta_a, \theta_b) = \cos\theta_a\cos\theta_b + \sin\theta_a\sin\theta_b = \cos(\theta_a - \theta_b), \tag{1.21}$$

which is (1.18). Equivalently, the probability that Alice and Bob obtain the same outcome is $\cos^2\frac{\theta_a-\theta_b}{2}$ and the probability that they differ is $\sin^2\frac{\theta_a-\theta_b}{2}$. With the settings above, $\theta_a = 0$,

$\theta_{a'} = \pi/2$, $\theta_b = \pi/4$, $\theta_{b'} = -\pi/4$, the three pairs (a, b), (a, b'), (a', b) are separated by an angle of $\pi/4$ in modulus and the pair (a', b') by $3\pi/4$, so

$$\begin{aligned} E(a,b) &= \cos\tfrac{\pi}{4} = \tfrac{1}{\sqrt{2}}, & E(a,b') &= \cos\tfrac{\pi}{4} = \tfrac{1}{\sqrt{2}}, \\ E(a',b) &= \cos\tfrac{\pi}{4} = \tfrac{1}{\sqrt{2}}, & E(a',b') &= \cos\tfrac{3\pi}{4} = -\tfrac{1}{\sqrt{2}}. \end{aligned} \tag{1.22}$$

The one negative correlator is exactly the one that enters (1.15) with a minus sign, so all four terms add constructively:

$$S = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = 2\sqrt{2} > 2. \tag{1.23}$$

Many texts use the singlet $|\Psi^-\rangle$ instead. Since $(Z \otimes Z)\,|\Psi^-\rangle = -\,|\Psi^-\rangle$ and $(X \otimes X)\,|\Psi^-\rangle = -\,|\Psi^-\rangle$, the same computation gives $E(\theta_a, \theta_b) = -\cos(\theta_a - \theta_b)$ for the singlet: perfect anticorrelation when the settings coincide, and $S = -2\sqrt{2}$ for the angles above. The magnitude of the violation is the same, and only the sign convention differs. Throughout this course we use $|\Phi^+\rangle$ and the correlator (1.18).

Remark 1.2 Scope of the violation

A measured value of S above 2 rules out every model satisfying the three assumptions of Definition 1.4 at once. It does not tell us which assumption fails; the standard reading gives up the joint assumption of locality and predetermined outcomes, but the theorem itself is silent on the choice. It does not permit signalling: each of the four correlators is compatible with Proposition 1.1, and Bob's outcomes remain a fair coin whatever Alice does. It does not mean that entanglement always violates a Bell inequality: measured only in the computational basis, $|\Phi^+\rangle$ gives $E = 1$ and is reproduced exactly by the coin-flip mixture, and the same is true of any settings taken from a single basis. The violation requires each party to choose between incompatible measurements. Finally, the bound is a mathematical theorem about a class of models; the claim that nature violates it is empirical. Experiments since the 1970s have observed violations, and since 2015 experiments have done so with the principal loopholes (detection efficiency and spacelike separation of the setting choices) closed simultaneously [3, 4, 5].

1.4 Summary

The state of two qubits is a unit vector in $\mathbb{C}^4$, built from the single-qubit spaces by the tensor product. A product state assigns a state to each qubit; an entangled state does not, and $|\Phi^+\rangle$ is the standard example. Two gates, H on one qubit and CNOT, produce it from $|00\rangle$. An n-qubit register has 2^n amplitudes, but a measurement returns n bits and none of the amplitudes directly. Measuring one qubit of an entangled pair leaves the other in a conditional state that depends on the outcome, in every basis; matching classical bits agree in one basis only. Because the outcome is random and Bob's marginal distribution is unchanged, no message travels. The CHSH inequality bounds the correlations of any local hidden-variable model by 2; $|\Phi^+\rangle$ with suitably chosen settings gives $2\sqrt{2}$. The violation rules out that class of models, and nothing more: not signalling, not communication, and not a violation for every choice of settings.

Homework 1, part B, can be attempted after this lecture.

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